1. SYSTEM IDENTIFICATION

1.1. System Name/Title: State Institute of Science and Technology (SIST) Web Infrastructure System (SWI)

1.1.1. System Categorization:
Confidentiality: High
Integrity: High
Availability: High
Overall Security Categorization: High Impact System

The SIST Web Infrastructure System is categorized as High impact based on the high-water-mark approach. The assessment identified High-impact information types within the AWS-hosted HR, GMAT, and FCIN applications. The system-level impact ratings reflect the highest applicable confidentiality, integrity, and availability ratings documented in the security categorization assessment.

1.1.2. System Unique Identifier: SIST-SWI-AWS-001

1.2. General Description/Purpose of System: What is the function/purpose of the system?
The SIST Web Infrastructure System is a simulated government information system supporting Human Resources (HR), Graduate Management Admission Testing (GMAT), and Financial and Contract Information Network (FCIN) applications.

The AWS-hosted environment includes web, application, and database servers running on Amazon EC2 instances. The proposed architecture uses AWS IAM for identity and access management, AWS CloudTrail for supported API activity logging, Amazon CloudWatch Logs for configured system and application logging, and Amazon S3 for audit log storage.

The system is designed to support the confidentiality, integrity, and availability of sensitive organizational information. These components represent the simulated architecture used for academic RMF assessment activities and do not constitute a verified production deployment.

1.2.1. Number of end users and privileged users:
Total users : 470
Privileged users: 20

Roles of Users and Number of Each Type:

Roles	Number of Users	Number of Administrators/ Privileged Users
Students	150	0
System administrators	5	5
Database administrators	3	3
Application administrators	10	10
Security administrators	2	2
General employees	300	0

1.3. General Description of Information: CUI information types processed, stored, or transmitted by the system are determined and documented. For more information, see the CUI Registry at https://www.archives.gov/cui/registry/category-list.

The simulated SIST Web Infrastructure System processes, stores, and transmits sensitive organizational information, including:

- Employee personal information and HR records
- Student records
- Financial and contract information
- System audit logs and security monitoring information

Some information may qualify as Controlled Unclassified Information (CUI), depending on applicable federal laws, regulations, and government-wide policies. Specific CUI categories would need to be identified and validated as part of the information classification process.

2. SYSTEM ENVIRONMENT

Include a detailed topology narrative and graphic that clearly depicts the system boundaries, system interconnections, and key devices. (Note**:** *this does not require depicting every workstation or desktop*, but include an instance for each operating system in use, an instance for portable components (if applicable), all virtual and physical servers (e.g., file, print, web, database, application), as well as any networked workstations (e.g., Unix, Windows, Mac, Linux), firewalls, routers, switches, copiers, printers, lab equipment, handhelds). If components of other systems that interconnect/interface with this system need to be shown on the diagram, denote the system boundaries by referencing the security plans or names and owners of the other system(s) in the diagram.

> The proposed SIST AWS Web Infrastructure System operates within an Amazon Virtual Private Cloud (VPC). Users access the web application through HTTPS. An Application Load Balancer distributes incoming requests to the web tier, which communicates with the application tier. The application tier accesses the database server running Red Hat Enterprise Linux. AWS CloudTrail records supported AWS API activity, while Amazon CloudWatch Logs collects configured system and application logs. Amazon S3 provides storage for audit logs.

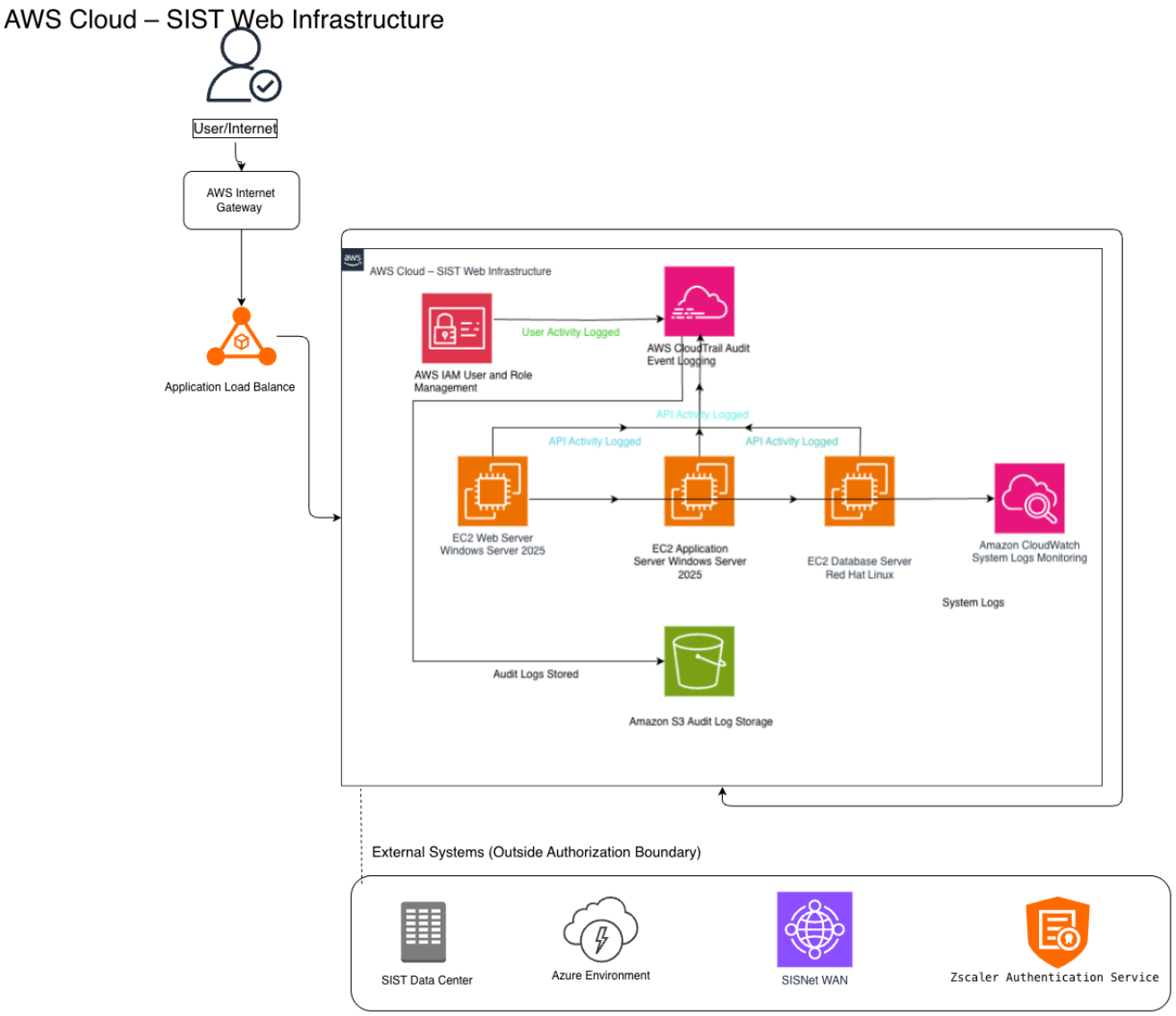


Figure 1. Proposed SIST AWS Web Infrastructure Architecture (Initial State).

2.1. Hardware and Infrastructure Inventory.

- Hardware resources are provided through AWS. The following table summarizes the proposed AWS infrastructure components included in the simulated assessment.

Hardware Component	**Vendor**	**Model/Service**	**Responsible Role**
Computer Infrastructure	AWS	Amazon EC2	Cloud Administrator
Storage Infrastructure	AWS	Amazon S3	Cloud Administrator
Networking Infrastructure	AWS	AWS VPC	Network Administrator

2.2. Software and AWS Services Inventory.

Software	Version	Purpose
Windows Server	2025	Web and Application Servers
Red Hat Enterprise Linux	9	Database server
Oracle Database	19c	Database management
Java Runtime Environment	17	Application runtime
AWS CloudTrail	Managed service	API audit logging
Amazon CloudWatch Logs	Managed service	System and application log monitoring
Amazon S3	Managed service	Audit log storage

2.3. Hardware and Software Maintenance and Ownership - Is all hardware and software maintained and owned by the organization?

AWS owns and maintains the underlying physical cloud infrastructure. SIST is responsible for its assigned cloud security responsibilities, including guest operating systems, application configurations, identity and access management, and organizational data. Responsibility for individual controls depends on the AWS service and applicable shared responsibility arrangements.